Topological Limits of Behavioral Regulation in Antitrust

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VERIFIED STATUS
Both complementary Lean developments built successfully. The final order-60 theorems report no `sorryAx`. All 21 independent search-tree certificates and all 35,967 nodes replayed successfully; the corrected integrity manifest passed in full.

1. Result

There is an explicit finite graph X on 60 vertices such that X is 4-vertex-critical and deleting any single edge leaves its chromatic number equal to 4. Equivalently, X has no critical edges. The formal structured conclusion is:

```
Dirac60.order60_solution :
  FourVertexCritical G /\ EdgeImmuneAtFour G
```

The compact development separately proves that X is not 3-colourable, gives a proper 4-colouring, checks a proper 3-colouring after every vertex deletion, and proves that no 3-colour assignment has at most one monochromatic edge.

2. Explicit graph

Let Gamma have underlying set Z/12Z x Z/5Z with multiplication

```
(e,i)(f,j) = (e+f mod 12, i + 2^e j mod 5).

S = {(1,0),(11,0),(5,1),(7,2),(6,0),(6,3)}.
X = Cay(Gamma,S),    label(e,i) = 5e+i.
N(0) = {5,26,30,33,37,55}.
```

This is a finite simple vertex-transitive 6-regular graph with 60 vertices and 180 edges. Its frozen edge-list digest is:

```
85914250312a4fe8d1a67d747fea2ee1772c9762796af90095efa61fb4c0400c
```

3. Why the certificate proves the graph result

Balanced punctures

Every proper 3-colouring of X-0 assigns each of the three colours to exactly two vertices of N(0). A displayed proper colouring of X-0 exists. By Cayley transitivity, the same statement holds at every deleted vertex.

Chromatic number and vertex criticality

A proper 3-colouring of X would restrict to a punctured colouring, but the colour assigned to the puncture already appears on two neighbours. Hence X is not 3-colourable. Giving the puncture a fresh fourth colour yields a proper 4-colouring, so chi(X)=4. Translated punctured witnesses show chi(X-v)=3 for every vertex v.

Edge immunity

If deleting an incident edge made the graph 3-colourable, at most one of the two same-colour neighbours supplied by the balanced-puncture property could be removed. A second monochromatic incident edge would remain. Transitivity covers every edge, proving chi(X-e)=4 for every edge e.

4. Two complementary Lean certificates

Route	Checked content	Outcome
Compact Boolean/SAT	Fully unrolled 120-bit colour encoding; q3(X)>=2; 60 deletion witnesses; proper 4-colouring.	Build and executable passed
Structured search tree	Generic soundness theorem; balanced-puncture transport; 21 explicit certificate trees.	Build, axiom audit, and replay passed

5. External Lean verification record

Compact route

```
Build completed successfully.
Dirac60 Lean certificate elaborated successfully.
edges: 180
all concrete vertex witnesses valid: true
displayed three-colour defect: 2
```

Structured route

The structured project built successfully. Its independent replay passed 21 cases and 35,967 nodes. The replay reproduced the frozen edge hash and certificate hash:

```
EDGE_SHA256 85914250312a4fe8d1a67d747fea2ee1772c9762796af90095efa61fb4c0400c
CERT_SHA256 5959d3f0943bba98e2278ae567f7bbe2a406e41f7ce2015fa6e0101b9449e195
```

Axiom audit

```
BalancedPuncture.twoPerColour_edgeImmune
  [propext]
Dirac60.valid_no_extension
  [propext, Quot.sound]
Dirac60.exact_two_two_two_profile
  [propext, Classical.choice, Lean.ofReduceBool, Quot.sound]
Dirac60.order60_solution
  [propext, Classical.choice, Lean.ofReduceBool, Quot.sound]
UniversalDecoupling.iff_ge_four
  [propext, Quot.sound]
```

No reported theorem depends on sorryAx. Lean.ofReduceBool records the practical trust in Lean's native evaluator/code-generation path used for concrete finite certificate replay.

6. Universal Decoupling Theorem

For k>=2, let D(k) mean that there exists a finite simple graph H with chi(H)=k, chi(H-v)=k-1 for every vertex v, and chi(H-e)=k for every edge e. Then:

> For every integer k>=2: D(k) if and only if k>=4.

Necessity: a 2-vertex-critical graph is K2, whose edge is critical; a 3-vertex-critical graph is an odd cycle, and deleting any cycle edge lowers its chromatic number. Sufficiency at k=4 is the order-60 graph above. Sufficiency for every k>=5 follows from Jensen's published theorem.

Formal scope

The Lean declaration UniversalDecoupling.iff_ge_four verifies the final logical deduction when supplied with three explicit hypotheses: the order-4 construction, Jensen's k>=5 existence input, and the low-chromatic obstruction. The order-60 input is formalized here. Jensen's infinite family is cited but is not itself formalized, so the universal theorem is not an end-to-end Lean theorem independent of the literature.

Logical qualification

This is existential by chromatic level. It does not assert that every vertex-critical graph with chromatic number at least four is edge-immune: K4 is an immediate counterexample to that pointwise reading.

7. Reproduction commands

```
# Compact route
cd lean_bv
lake update
lake exe cache get
lake build
lake exe dirac60-check

# Independent structured route
cd ../lean_search_tree
lake update
lake exe cache get
lake build
lake env lean AxiomAudit.lean
python3 replay_certificates.py
shasum -a 256 -c SHA256SUMS
```

8. References and acceptance boundary

T. R. Jensen, Dense critical and vertex-critical graphs, Discrete Mathematics 258 (2002), 63-84. E. Skottova and R. Steiner, Critical edge sets in vertex-critical graphs, arXiv:2508.08703. A. Ferudun, Exact 6-cut rigidity and small-order superconnectivity for the 6-regular case of Dirac's k=4 problem, arXiv:2606.18462.

The order-60 statement is Lean-checked within the axiom boundary printed above. Independent public reproduction, expert inspection of the definitions and certificates, and peer review remain necessary before treating the result as accepted in the mathematical literature.